Problem set 1

Mathematical background / Euclidean space and motions

Modern Robotics

2013

1. Coordinate transformations

Consider an object moving with a speed **v**, on which a force **F** is applied in the opposite direction (this could for instance be a friction force). **v** and **F** are depicted in Figure 1.

The velocity can be expressed as an array of numbers using the indicated coordinate frame $\bar{\Psi}$. For simplicity, the force **F** is also expressed as a row vector (array of numbers) $\bar{F}$, but beware that it is a co-vector and not a vector:

$$\bar{v} = 2\begin{pmatrix}\cos\alpha \\ \sin\alpha\end{pmatrix} \qquad \bar{F} = \begin{pmatrix}-\cos\alpha \\ -\sin\alpha\end{pmatrix} \tag{1}$$

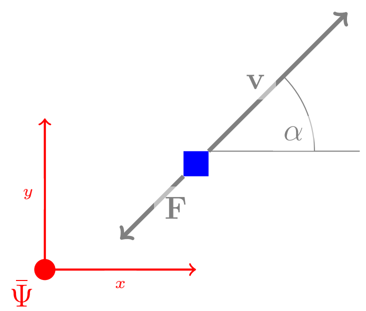

Figure 1: A moving object with a force acting on it. Note that, while both **F** and v are depicted using arrows, only one is a vector whilst the other is a co-vector!

Consider a second, conveniently oriented frame Ψ and calculate the matrix A that maps $\bar{v}$ into the new frame, i.e.

$$v = A\bar{v} \tag{2}$$

Then show how this transformation holds for F vs. $\bar{F}$. (Hint: you can find this transformation by using power-continuity in both frames, i.e. "$P = \bar{P}$".)

Taking a frame that has its x axis aligned with **v**, so ψ is rotated with an angle of α w.r.t. $\bar{\psi}$, we find that A is the well-known 2-D rotation matrix:

$$A = \begin{bmatrix}\cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha)\end{bmatrix}$$

Now $\mathbf{v}$ is a one-dimensional array:

$$v := A\bar{v} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix} \cdot 2 \begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

To find the mapping A^* for the force, which is a co-vector, we use power-continuity—the power expressed in frame ψ should be equal to the power in frame $\bar{\psi}$:

$$v = A\bar{v}, \quad F = A^*\bar{F}$$

$$\begin{aligned}
F^\top v &= \bar{F}^\top \bar{v} && \text{fill in } v = A\bar{v} \\
F^\top A\bar{v} &= \bar{F}^\top \bar{v} && \text{Should hold for all } \bar{v} \\
F^\top A &= \bar{F}^\top && \text{Take transpose left and right } ((MN)^\top = N^\top M^\top) \\
A^\top F &= \bar{F} && \text{Multiply with} (A^\top)^{-1} \\
F &= (A^\top)^{-1}\bar{F} && F = A^*\bar{F} \\
A^* &= A^{-\top}
\end{aligned}$$

Do you notice anything peculiar? For a 2×2 matrix, finding the inverse is easy: divide by the determinant, swap the diagonal elements and negate the off-diagonal terms. If you do this for A, you will find that $A^{-1} = A^\top$: a special property of orthonormal rotation matrices is that their inverse is their transpose. This also means that $A^* = (A^{-1})^\top = (A^\top)^\top = A$; in other words: in this special case the mapping for vector v and covector F was the same.[1]

[1]Later on, when considering full 6-D motions/transformations, we *will* get different mappings for forces and velocities.

2. Tilde form

The tilde form as introduced in the lectures is a 3×3 skew-symmetric matrix for which holds (with v and w vectors of length 3):

$$v \wedge w = \tilde{v} w \tag{3}$$

Why does this only work in a 3-dimensional world?[2]

For a skew-symmetric matrix, $M^\top = -M$. Any element on the diagonal, m_{ii}, will not move after taking the transpose, which means that $m_{ii} = m_{ii}^\top$, but from $M^\top = -M$ we find that $m_{ii} = -m_{ii}^\top$. This means that $m_{ii} \equiv 0$: all diagonal elements of a skew-symmetric matrix are zero.

All non-diagonal elements swap places in pairs when taking the transpose, and from the definition $M^\top = -M$ we see that $m_{ij} = -m_{ji}$: the off-diagonal elements of M are in pairs that are eachother's negative.

For an $n \times n$ skew-symmetric matrix, this means that the number of independent elements is n^2 (number of elements) minus n (diagonal), divided by 2: $n_{\text{free}} = (n^2 - n)/2$. If this $n \times n$ matrix must have a unique relation to an n-dimensional vector, the number of free elements in the matrix must be n:

$$n = n_{\text{free}} = (n^2 - n)/2$$

If you solve this equation, you'll find $n = 3$.

3. Skew-symmetry of $\tilde{\omega}$

According to the theory, the transport of the vector $\dot{R}$ back to the identity (by applying $R^\top$) yields $\tilde{\omega}$, a skew-symmetric matrix. Prove that $R^\top \dot{R}$ is indeed skew-symmetric.

$$\begin{aligned}
R^\top &= R^{-1} && \text{orthonormal matrix, definition of SO(3)} \\
RR^\top &= RR^{-1} = I_3 && \text{matrix times its inverse is identity} \\
\dot{RR^\top} &= \dot{I}_3 = 0 && \text{Time derivative; dot is on the pair } RR^\top \\
(R\dot{R}^\top) + (\dot{R}R^\top) &= 0 && \text{product rule of derivative} \\
(R\dot{R}^\top) &= -(\dot{R}R^\top) && \text{Now apply to the right } (NM^\top) = (MN^\top)^\top \\
(R\dot{R}^\top) &= -(R\dot{R}^\top)^\top && \text{Definition of s.s. matrix: } M = -M^\top
\end{aligned}$$

[2] Actually, in higher dimensions you can still make it work, but there the bivectors associated with the skew-symmetric matrix have redundant degrees of freedom, so they are not uniquely related to the vectors.

4. Composition of H-matrices

Consider three 2D coordinate frames Ψ_1,Ψ_2 and Ψ_3. The coordinate transformation from Ψ_1 to Ψ_2 consists of a rotation of 30°, and the coordinate transformation from Ψ_2 to Ψ_3 consists of a translation of $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$. Derive the 2D homogeneous matrices H_2^1 and H_3^2 representing these transformations, as well as H_3^1, the composition of these two transformations.

We want to find H_3^1, to be able to transform points that are expressed in Ψ_3 to points expressed in Ψ_1: $P^1 = H_3^1 P^3$. Working backwards in steps from 3 to 1, going by 2:

$$H_3^2 = \begin{bmatrix} \mathbf{R_3^2} & \mathbf{o_3^2} \\ \mathbf{0}^\top & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \qquad \text{Origin of 3 in frame 2, } \mathbf{o_3^2} easily found.$$

$$H_2^1 = \begin{bmatrix} \mathbf{R_2^1} & \mathbf{o_2^1} \\ \mathbf{0}^\top & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}\sqrt{3} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2}\sqrt{3} & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \text{Only a rotation matrix between 2 and 1.}$$

Now it is straightforward to find H_3^1:

$$H_3^1 = H_2^1 H_3^2 = \begin{bmatrix} \frac{1}{2}\sqrt{3} & -\frac{1}{2} & \sim 0.10 \\ \frac{1}{2} & \frac{1}{2}\sqrt{3} & \sim 5.83 \\ 0 & 0 & 1 \end{bmatrix}$$